

LESSON 3: SYSTEMS & INTERFACES

The concept of a system is fundamental to our understanding of ourselves and the world around us. In this lesson, we will explore computers as technological systems and relate them to humans as biological systems, with the commonality being that they are both capable of information processing. The interaction that occurs between computers and humans underlies the technological developments that have led to the tremendous evolution in our way of life.

SYSTEMS & SIGNALS

A system is a set of interconnected components that work together to achieve a particular task or purpose. These parts or components interact with each other in a coordinated manner, allowing the system to function as a cohesive whole.

A SYSTEM IS A SET OF INTERCONNECTED COMPONENTS THAT WORK TOGETHER AS A WHOLE, SERVING A PARTICULAR PURPOSE.

In order for a system to work as a whole, all its parts must interact with each other and coordinate their functions and activities. In the case of systems that depend on information processing, such as humans and computers, this means that the individual parts of a system must be able to communicate with each other; they must be able to exchange information.

Notes

Humans are one example of a biological system capable of information processing. In fact, any living organism with a central nervous system qualifies to be described as an information processing biological system.

A signal is a message that is carried in physical form; it is essentially the transfer of energy in a fashion that carries some meaningful content interpretable by some information processing system.

A SIGNAL IS A MESSAGE CARRIED IN PHYSICAL FORM.

In computers, the form of energy that typically acts as a signal is electrical energy; more precisely, it is electrical potential energy. Humans also use electrical potential energy for internal signaling. In both cases, changes in voltage are propagated along pathways that connect the various parts of the system. For computers, these signals are called digital electronic signals, and they are propagated along conducting wires. Whereas for humans, these signals are called action potentials, and they are propagated along nerves.



Action potential
signals



Digital electronic
signals

Signals play a crucial role in the coordination of the internal components of a system, as they enable communication between different parts and ensure proper functioning.

**SIGNALS ENSURE COMMUNICATION AND COORDINATION BETWEEN
THE PARTS OF A SYSTEM.**

INTERFACES & TRANSLATION

The boundary of a system where it interacts with the outside world is called its interface. For information processing systems, interaction entails an exchange of signals. Signals going into the system are referred to as input, and in response, the signals that come out are output.

THE INTERFACE IS THE BOUNDARY AT WHICH A SYSTEM INTERACTS WITH THE EXTERNAL WORLD.

Every information processing system has its own form of internal signals; humans use action potentials, and computers use digital electronic signals. These are the only signals they can use to communicate and process internally.

External signals of the outside world are quite different from internal signals; they must be converted into internal signals. Light, sound, pressure and any other signal of the outside world must be converted into either action potentials or digital electronic signals. This conversion from one form of signal to another must preserve the integrity of the message itself. In a sense, it is quite analogous to translation between languages, wherein the information conveyed in one language must be preserved, but the language in which it is conveyed is changed. Signal conversion occurs at the interface.

SIGNAL CONVERSION OCCURS AT THE INTERFACE.

INPUT DEVICES

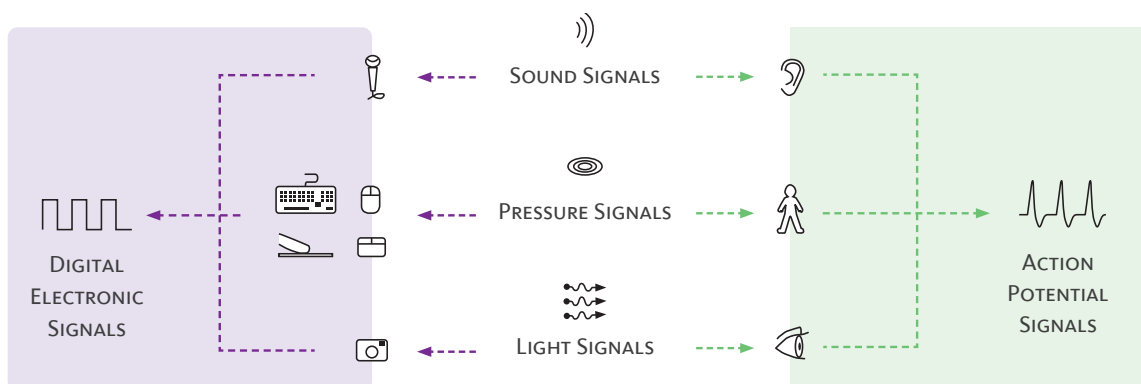
Input devices are the parts of the system that handle information coming in. External signals are converted to internal signals while preserving the infor-

mation they carry. This ensures that the system gets accurate information about its surroundings. Each device is dedicated to converting a specific type of external signal into the required internal signal.

THE INPUT DEVICE CONVERTS A SPECIFIC FORM OF EXTERNAL SIGNAL INTO THE INTERNAL SIGNAL OF THE SYSTEM.

Computer parts responsible for input are called input devices and are often referred to as sensors. Each device is dedicated to the task of converting one form of signal into electronic digital signals. Humans, similarly, have sensory organs which are responsible for accepting input and converting one form of external signal into internal action potential signals.

Sound, pressure and light are commonly converted external signals by both computers and humans. Computers have a microphone and camera to convert sound and light, respectively, while humans have ears and eyes. Computers have various input devices that are capable of converting pressure depending on the type of computer; these are the keyboard, mouse, touchpad and touchscreen. Humans have the skin as a sensory organ capable of converting pressure signals into action potentials.



Sound, pressure and light are commonly converted external signals by both computers and humans. The computer has various input devices, such as the microphone, camera, keyboard, mouse, touchpad and touchscreen, to convert sound, light and pressure into internal signals. In comparison, the human body has the ear, eye and skin, respectively.

In addition, many smartphones are equipped with input devices or sensors that can detect position and motion. Some of the most common sensors used for this purpose include:

1. **Accelerometers** are used to detect linear motion in any of the 3 dimensions: up or down, left or right and forward or backward.
2. **Gyroscopes** are used to detect rotational motion in any one of the 3 dimensions: rotating clockwise or counterclockwise, rolling left or right and pitching up or down.
3. **Magnetometers** are used for magnetic fields. They are used to detect the orientation of the smartphone relative to the Earth's magnetic field.
4. **GPS** are sensors that use satellite signals to determine the location of its computer or device on Earth.

These four sensors work together to determine speed, direction, orientation, position and location, allowing the smartphone to track its own movement and its relative position in Space. This makes them useful in a variety of applications, such as games, fitness trackers, maps and navigation.

In comparison, while humans do not have sensory organs that can detect the Earth's magnetic field or GPS satellite signals, their vestibular system provides highly accurate information about linear movement, rotational movement, orientation and position in space. We can easily tell if we are moving and in which direction. We can also easily tell if we are leaning in any direction relative to the ground.

Furthermore, humans have additional senses. The skin not only provides pressure information, but it also provides information about temperature and pain. In addition, humans are capable of detecting chemical signals through taste and smell.

OUTPUT DEVICES

In contrast to input devices, output devices are responsible for responding to the outside world. Responses, or output, can be in the form of a signal or an action, each of which requires a dedicated output device. In either case, an internal electronic digital signal is converted to an external signal or an external action. This is similar to the case of humans in whom an internal action potential is converted to an external signal, such as the sound of speech, or an external action, such as the movement of a limb.

THE **OUTPUT DEVICE** CONVERTS THE INTERNAL SIGNAL OF THE SYSTEM TO A SPECIFIC FORM OF EXTERNAL SIGNAL OR ACTION.

Computers mainly output two forms of signals: light from the screen display and sound from the speaker. In addition, the handheld computers, namely the tablet and smartphone, can also vibrate, emitting what is called a haptic signal or pressure that can be felt by whoever is holding the computer.

Output devices that respond by performing an action are called actuators; they are capable of moving. Actuators are part of other machines or devices that can be connected to a computer or are controlled by an embedded computer. The printer is a machine that connects to a computer and contains an actuator. The printer's printhead is an actuator that moves across the paper, printing line by line. Other examples of actuators include motors and valves, which are used in various machines such as home appliances, vehicles, manufacturing equipment and robots.

THE **ACTUATOR** IS AN OUTPUT DEVICE CAPABLE OF PERFORMING AN ACTION.



HARDWARE & SOFTWARE

The human computer interface is not just a physical one: it is also a logical one. The hardware component of the interface includes the various physical components, the input and output devices, which are responsible for transmitting and receiving signals. The hardware of the computer handles the physical layer of the interface, which are the signals it is transmitting and receiving.

THE PHYSICAL LAYER IS THE SIGNALS; IT IS HANDLED BY THE HARDWARE.

The software component of the interface determines the information (logic) to be carried by the signals. The software component includes the applications and other programs that determine how the computer behaves and what information it shows the user. The content displayed on the screen is determined by software. What happens when a user clicks a button is determined by software. Everything a computer does is determined by software.

So when we interact with computers, we are interacting with both hardware and software. At the human computer interface, an exchange of signals carrying information is occurring, with the signals being the physical layer and the information being the logical layer.

THE LOGICAL LAYER IS THE INFORMATION IN THE SIGNALS; IT IS HANDLED BY THE SOFTWARE.

EXAMPLE 3.1

When two humans have a conversation, they are interacting. This interaction occurs in both physical and logical layers. In the physical layer, there is an exchange of sound signals, which are sensed by the ears and emitted by the mouth. However, the content of the conversation, the actual information represented by these sound signals, is what forms the logical layer. The logical layer contains the meaning carried by the sound signals.

USER INTERFACE

The user interface (UI) is a term that refers to the software component of the human computer interface, also referred to as the software interface. The UI is responsible for translating user input into instructions that the computer can understand and execute, as well as displaying the results of those instructions for the user.

THE USER INTERFACE IS THE SOFTWARE COMPONENT OF THE HUMAN COMPUTER INTERFACE.

There are several types of user interfaces that evolved over time; these include in chronological order:

1. **Command-Line Interface (CLI):** A CLI requires users to type commands into a text-based interface to perform functions. CLI interfaces are commonly used in programming and system administration but are challenging for users unfamiliar with programming or technical jargon. Examples include Windows Command Prompt and macOS Terminal.

```
Last login: Wed Apr 13 15:35:52 on ttys001
computer-name:~ username$
```

The Terminal window on macOS showing a standard output typical display. The first line indicates the date and time of the last login. The second line displays the name of the computer (in this case, “computer-name”) followed by the username (in this case, “username”) and a dollar sign (\$), indicating that the terminal is ready to accept commands.

- 2. Graphical User Interface (GUI):** A GUI uses graphical elements such as windows, icons and menus to represent different functions and commands. The GUI made computer systems more accessible and user-friendly by reducing the need for users to remember complex commands and by providing a more intuitive and visual interface. Examples include Windows and macOS.

Graphical Elements of the GUI

The GUI has three common graphical elements: windows, icons and menus. A window is a separate area of the screen containing content related to a specific task or application. More than one window can be opened, allowing users to interact with multiple applications or tasks simultaneously. In addition, icons and menus allow users to select applications, files and commands.

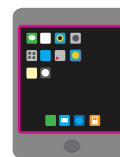
THE GRAPHICAL USER INTERFACE USES WINDOWS, ICONS AND MENUS TO REPRESENT DIFFERENT FUNCTIONS AND COMMANDS.



Desktop



Laptop



Tablet



Smartphone

Desktops and laptops have pointing devices, the mouse and touchpad, respectively. These hardware components are used to control a software component displayed on the screen, the pointer. Tablets and smartphones have touchscreens, a technology that allows pointing to be replaced by touching.

There are two main types of GUIs that are distinguished based on the technology they incorporate to select and control graphical elements: the WIMP and touchscreen interfaces.

WIMP Interface: The acronym WIMP stands for windows, icons, menus and pointer. A pointer is a graphical element, typically in the form of an arrow or hand, used to interact with the interface by selecting icons and items in menus and windows. It is controlled by a mouse or touchpad, both of which are input devices known as pointing devices.

Touchscreen Interface: With the rise of mobile devices, the touchscreen interface became popular, allowing users to interact with devices directly by touching the screen using natural gestures such as swiping, tapping, or pinching, making it easy for users to learn and use. Touchscreens kept the windows, icons and menus but eliminated the need for pointing devices, such as the mouse or keyboard.

3. **Voice User Interface (VUI):** Most recently, the (VUI) has become more prevalent, allowing users to interact with devices using natural language and voice commands. VUIs are becoming increasingly popular in smart home devices and personal computers, such as Apple's Siri, Amazon's Alexa, Google's Assistant and Microsoft's Cortana.

Career Tips

Human Computer Interaction is a multidisciplinary field that combines principles from computer science, cognitive psychology, design and other fields to study how people interact with computers and how to best design user interfaces.

USABILITY

Usability is a design feature that refers to how easy it is for users to interact with a computer in order to complete a particular task. The evolution of the user interface and its related hardware components reflects an evolution in usability.

USABILITY IS THE EASE WITH WHICH A DEVICE CAN BE USED TO COMPLETE A PARTICULAR TASK.

The touchscreen interface is often considered highly intuitive; users can simply touch the screen to interact with it, which many people find easy to learn and use regardless of their knowledge and skills with computers.

The WIMP interface requires a mouse or touchpad, which may require some learning and practice for users who are unfamiliar with the technology. However, once users become familiar with the interface, it can be quite efficient and effective for many computing tasks.

The CLI requires more learning and memorization of specific commands on the part of the user. As a result, it is generally only usable by IT professionals or those with advanced computing skills.

The VUI is an interface that has emerged in recent years. It allows users to interact with computers through speech. While it may seem a step up in usability, this interface is still in its early stages of development.

Overall, the usability of a user interface is an important consideration in designing effective and efficient human-computer interaction. The evolution of hardware components and the user interface itself has reflected a trend towards more intuitive and easy-to-use interfaces, which has allowed a wide range of people to use computers.

SUMMARY

A system is a set of interconnected components that work together as a whole, serving a particular purpose. Humans and computers are examples of information processing systems.

A signal is a message carried in physical form. In an information processing system, signals ensure communication and coordination between the parts of the system. Every system has its own specific type of internal signal, e.g. humans have action potentials, and computers have digital electronic signals.

The interface is the boundary at which a system interacts with other systems and the external world. Signals must be converted between external signals and internal signals. This conversion occurs at the interface and must preserve the integrity of the information carried by the signals.

The interface of a system comprises both input and output devices. Input devices convert external signals into internal signals. Output devices convert internal signals into external signals. Each device is dedicated to a specific type of external signal.

Some output is in the form of action. The output device that performs this action is referred to as an actuator.

Information exists in two layers: the physical and virtual layers. The physical layer contains the actual signals and is handled by the hardware. The virtual layer is the information represented in the signals and is handled by the software.

The user interface is the software component of the human computer interface. This interface has evolved from a command-line interface to a graphical user interface and, more recently, a voice user interface.

The graphical user interface uses windows, icons and menus to represent different functions and commands. The WIMP interface and touchscreen interface are both types of graphical user interfaces. They both have windows, icons and menus, but the WIMP interface includes either a mouse or touchpad, which the touchscreen interface replaces with a touchscreen.

Usability is the ease with which a device can be used to complete a particular task.

A directory is a logical construct used by the filing system to organize files in a computer. A directory is used to hold files and, in many cases, other directories, referred to as subdirectories. The term directory is synonymous with “folder”.

Activity 3.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u> T </u>	A system is a set of interconnected components that work together to achieve a particular task or purpose.
<u> F </u>	Humans and computers do not share any common features as systems.
<u> F </u>	Signal conversion at the interface does not need to preserve the integrity of the message.
<u> T </u>	Input devices convert external signals into internal signals while preserving the information they carry.
<u> T </u>	Both humans and computers are capable of converting sound, pressure and light signals.
<u> F </u>	Actuators are only used in computers and not in other machines or devices.
<u> F </u>	The human-computer interface only includes hardware components.
<u> F </u>	Command-Line Interface (CLI) is the most user-friendly type of interface.
<u> T </u>	The Graphical User Interface (GUI) made computers more accessible by providing a more intuitive and visual interface.
<u> F </u>	The Voice User Interface (VUI) is a highly developed and mature interface technology.

Activity 3.2

Answer the questions below.

1. What do computers and humans have in common as systems?
 - a) They both use digital electronic signals.
 - b) **They are both capable of information processing.**
 - c) They both use action potentials.

2. What is the purpose of an interface in an information processing system?
 - a) To store information
 - b) To create new signals
 - c) **To convert external signals into internal signals**

3. Which form of energy typically acts as a signal in computers?
 - a) Thermal energy
 - b) **Electrical potential energy**
 - c) Kinetic energy

4. In the context of information processing systems, what is a signal?
 - a) A physical form of a message
 - b) A method of storing data
 - c) **A type of energy conversion**

5. Signals going into an information processing system are referred to as:
 - a) Output
 - b) **Input**
 - c) Interface

Activity 3.3

Answer the questions below.

1. What is the primary function of input devices?
 - a) Converting internal signals to external signals
 - b) **Converting external signals to internal signals**
 - c) Storing signals

2. Which two devices are responsible for converting sound signals?
 - a) Microphone - Keyboard
 - b) Mouse - Speaker
 - c) **Microphone - Speaker**

3. What type of sensor is used to detect linear motion in smartphones?
 - a) Gyroscope
 - b) **Accelerometer**
 - c) Magnetometer

4. What type of output device is responsible for performing an action?
 - a) Sensor
 - b) **Actuator**
 - c) Transducer

5. Which of the following is an example of an actuator in a printer?
 - a) Ink cartridge
 - b) Paper tray
 - c) **Printhead**

Activity 3.4

Answer the questions below.

1. The human-computer interface consists of:
 - a) Both input and output devices
 - b) Only input devices
 - c) Only output devices

2. Which type of interface uses elements such as windows, icons and menus?
 - a) Command-Line Interface
 - b) Voice User Interface
 - c) Graphical User Interface

3. Which interface allows users to interact with devices using natural gestures?
 - a) WIMP Interface
 - b) Touchscreen Interface
 - c) CLI Interface

4. The term “usability” refers to:
 - a) How easy it is for users to interact with a computer
 - b) The amount of memory a computer has
 - c) The processing power of a computer

5. Which interface is considered the most user-friendly for non-technical users?
 - a) Command-Line Interface (CLI)
 - b) Graphical User Interface (GUI)
 - c) Voice User Interface (VUI)

Activity 3.5

Answer the questions below about signal conversion at the interface.

1. What is the purpose of signal conversion at the interface of an information processing system?

The purpose of signal conversion at the interface is to transform external signals from the outside world into internal signals while preserving the integrity of the message. This conversion allows the system to use and respond to the information it receives from the external environment.

2. Why is preserving the integrity of the message important during signal conversion at the interface?

Preserving the integrity of the message during signal conversion at the interface is crucial because it ensures that the information conveyed by the signal remains accurate and meaningful when it is transformed into the internal signals used by the information processing system. This allows the system to correctly interpret and respond to the input it receives from the external environment.

Activity 3.6

Answer the questions below about humans and computers.

1. State the similarities that exist between humans and computers.

They are both systems.

They both can process information.

They both can sense and respond to their environment.

They are both capable of signal conversion at their interface.

They both use electrical potential energy for internal signaling.

2. “The ability to gather information from the environment adds functionality to a system and enhances its chances of success.”
Comment and give an example.

The more information a system can gather from its environment, the better

its chances of navigating through its environment to find sources of energy

and shelter, as well as to avoid obstacles and threats. Each sense adds a new

set of information. Therefore, adding senses or input devices adds more

information. The sense of smell can help someone avoid a fire before seeing it.

The sensing of Earth’s magnetic field helps birds migrate to warmer climates.

Having a smartphone with more sensors makes it more useful.

Activity 3.7

Answer the questions below about input and output devices.

1. What does the term “actuator” refer to in technology? Provide an example.

An actuator is an output device that converts internal digital electronic signals to movement. An example is the motor that drives the turntable of a microwave oven.

2. Identify and list the input devices on your personal computer.

Windows: 1) Right-click on “Start”, 2) select “Power User Menu”, 3) select “Device Manager” and then search the listed hardware categories.

macOS: 1) Click on the Apple logo, 2) select “About This Mac”, 3) select “System Report” or “More Info” and then search the listed hardware categories.

Androids & iPhones: A third-party app, such as “Sensors Multitool” or “Sensors Toolbox” will provide device-sensor information.

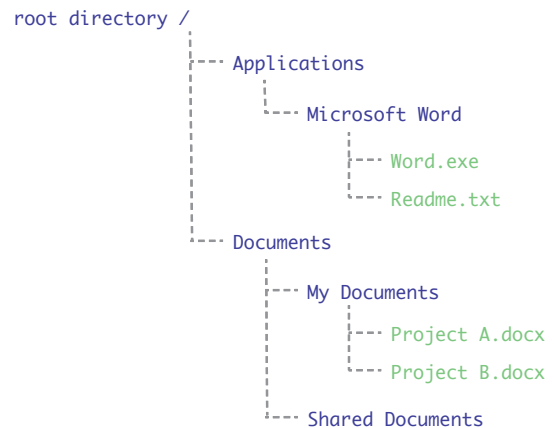
DIRECTORIES

Any information stored in a computer is stored in the form of a file. All files are organized and managed by the computer's filing system. The filing system uses the idea of a folder to organize files. A folder acts as a virtual container used to hold files and, in many cases, other folders, referred to as subfolders.

A more technical term for folder is directory. To restate using this term, the filing system organizes files under directories. A directory is used to hold files and, in many cases, other directories, referred to as subdirectories. This organization of directories within other directories is referred to as a hierarchical structure.

In this example, the directory structure shows the root directory and two levels of subdirectories.

In addition, four example files are shown that appear in green with their file extensions.



In the GUI, the folder icon is a graphical element of the user interface used to represent a directory. It is a visual abstraction of the more complex filing system managed by the computer. It is a simplification that allows a wide range of users to use the computer without having to deal with complex filing systems.

Notes

Abstraction is a fundamental concept in many fields, including computer science, sciences, mathematics and philosophy. It allows us to focus on the essential characteristics of a concept or system without getting bogged down in irrelevant details and is, therefore, a key aspect of problem-solving, innovation and creativity.

Activity 3.8

Use the command line interface to create a file in a specific folder. Then, find and add content to this file through the graphical user interface. Compare the two interfaces.

1. In the CLI, create a text file in the Documents directory and name it “My File” (see instructions). Add content, then save and close the file.
2. In the GUI, open “My File”, add additional content, then save and close the file.
3. Compare the two interfaces and their usability.

Instructions for macOS

1. Open Terminal, and list the directories under the root directory using the ‘ls’ command.

```
ls
```

2. Change directory using the ‘cd’ command followed by the name of the destination directory, which in this case is ‘Documents’.

```
cd 'Documents'
```

3. List the contents of the directory sorted in alphabetical order using the ‘ls | sort’ command.

```
ls | sort
```

4. Create a text file using the ‘touch’ command followed by the name of the file and its extension.

```
touch 'My File.txt'
```

5. Open the text file using the ‘nano’ command followed by the name of the file and its extension.

```
nano 'My File.txt'
```

6. Add text by simply typing the text, then save and close using ‘Ctrl+X’ and then ‘Y’ followed by ‘Enter’.

7. List the contents of the directory again to check that the file was created.

```
ls | sort
```

Instructions for Windows

1. Open Command Prompt, and list the directories under the root directory using the ‘dir’ command.

```
dir
```

2. Change directory using the ‘cd’ command followed by the name of the destination directory, which in this case is ‘Documents’.

```
cd "Documents"
```

3. List the contents of the directory sorted in alphabetical order using the ‘dir’ command with the ‘/O:N’ option.

```
dir /O:N
```

4. Create a text file using the ‘notepad’ command followed by the name of the file and its extension to open a file in the Notepad app.

```
notepad "My File.txt"
```

5. Add text by simply typing the text, then save and close using ‘Ctrl+S’ then ‘Ctrl+W’.

6. List the contents of the directory again to check that the file was created.

```
dir /O:N
```

Activity 3.9

Use the command line interface and the graphical user interface to create a backup folder that contains copies of your original files. Compare the two interfaces.

1. In the CLI, create a new directory under Documents. Name the new directory My Backups and move a copy of the file you created in Activity 3.8 into this directory.
2. In the GUI, create another directory under Documents. Name the new directory My Backups 2 and move a copy of the file you created in Activity 3.8 into this directory.
3. Compare the two interfaces and their usability.

Instructions for macOS

1. Open Terminal, and navigate from the root directory to the 'Documents' directory.

```
cd 'Documents'
```

2. Make a new directory using the 'mkdir' command. Name the directory 'My Backups'.

```
mkdir 'My Backups'
```

3. Copy the file 'My File.txt' using the 'cp' command. Name the copy 'Copy My File.txt'.

```
cp 'My File.txt' 'Copy My File.txt'
```

4. Move the copied file to the new directory 'My Backups'.

```
mv 'Copy My File.txt' 'My Backups'
```

5. Change directory using the 'cd' command to the 'My Backups' directory.

```
cd 'My Backups'
```

6. List the contents of the directory to check that the file is in it.

```
ls
```

Instructions for Windows

1. Open Command Prompt, and navigate from the root directory to the 'Documents' directory. Replace {your_username} with your actual username.

```
cd C:\Users{your_username}\Documents
```

2. Make a new directory using the 'mkdir' command. Name the directory 'My Backups'.

```
mkdir "My Backups"
```

3. Copy the file 'My File.txt' using the 'cp' command. Name the copy 'Copy My File.txt'.

```
copy "My File.txt" "Copy My File.txt"
```

4. Move the copied file to the new directory 'My Backups'.

```
move 'Copy My File.txt' 'My Backups'
```

5. Change directory using the 'cd' command to the 'My Backups' directory.

```
cd 'My Backups'
```

6. List the contents of the directory to check that the file is in it.

```
dir
```